

ATUL KUMAR MISHRA

Bengaluru, India

 atul-k-m —  atulkm189@gmail.com —  +91-9693235537 —  in LinkedIn

SUMMARY

Results-driven Information Science student with strong foundations in software engineering, object-oriented programming, data structures, and database systems. Passionate about building scalable systems, solving complex engineering problems, and delivering high-quality software in collaborative environments.

TECHNICAL SKILLS

Languages: Python, C++, C, JavaScript, SQL

Web Technologies: HTML, CSS, React, Next.js, Node.js, FastAPI

Tools & Platforms: Git, REST APIs, MongoDB, Excel

Core Areas: Data Structures & Algorithms, Object Oriented Programming (OOP), DBMS

PROJECTS

Sentinel IDS – ML-Based Intrusion Detection System

Python, Next.js, Machine Learning

- Designed and implemented a high-throughput network traffic analysis pipeline that achieved **97.5% detection accuracy** through rigorous model tuning and optimization.
- Designed ensemble anomaly detection models that reduced false positives by **21%**; applied root-cause analysis techniques to iteratively debug and improve detection logic.
- Implemented an explainable AI pipeline using SHAP, enabling transparent interpretation of model decisions and improving model interpretability, debugging, and validation workflows.
- Optimized the processing pipeline to handle **12,000+ flows/second**, demonstrating performance engineering and scalability best practices.

ReelAgent – Autonomous AI Video Production System

React, Python, FastAPI, Docker

- Built an autonomous end-to-end video generation pipeline using modular architecture and scalable software design principles.
- Designed an OOP-based microservice architecture with clean separation of concerns, enabling independent development, testing, and deployment of each component.
- Containerized deployment via Docker, ensuring reproducible builds, environment consistency, and scalable execution across platforms.

unipm – Universal Cross-Platform Package Manager

C++, Shell Scripting

- Designed and implemented a unified CLI tool applying graph-based dependency resolution and fuzzy-matching algorithms across 10+ package managers.
- Developed automated cross-platform installation workflows and shell integration scripts for Linux, macOS, and Windows environments.

LEADERSHIP EXPERIENCE

Technical Head – Entrepreneurship Cell (CodeRed 3.0 Hackathon)

2025

- Led a cross-functional technical team to develop and deploy the CodeRed'25 Hackathon platform, coordinating developers, testers, and infrastructure workflows across deployment cycles.

EDUCATION

BMS Institute of Technology and Management

Bengaluru, India

Bachelor of Engineering in Information Science and Engineering

2023 – 2027

CGPA: 8.55

RELEVANT COURSEWORK

Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, Software Engineering

ACHIEVEMENTS

- 3rd Place – BIZ TRIATHLON at ENIFINITY 2025, NMIT Bengaluru.